Part I: Full-Width Stream

Heading Huge (Full-Width)

Sub-Heading LARGE

Sub-Sub-Heading Large

Sub-Sub-Sub-Heading Large

Regular paragraphs are in normal size, indented with no vertical space between them. In this stream, the text spans the entire text area from the left blank margin to the right blank margin.

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Topic Environment Topic environments start with an unindented line where the topic name starts the line and the first paragraph continues after it inline. Subsequent paragraphs indent with no vertical space.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Sub-Topic Environment Subtopic environments are a regular indented paragraph same as a regular paragraph in topic, but start with a subtopic name. The first paragraph continues after the subtopic name inline. Subsequent paragraphs are indented.

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

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Sometimes math is inline:

- The scalar value $\alpha \in \mathbb{R}$ satisfies the condition $\alpha^2 > 0$ for all non-zero inputs.
- Let $f(x) = e^{-x}\sin(x)$ be a continuous mapping on the open interval $(0, 1)$.
- Given vectors $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (4, 5, 6)$, their dot product is $\mathbf{u} \cdot \mathbf{v} = 32$.

Other times it is a displayed derivation:

$$ax^2 + bx = -c$$

Divide both sides by a:

$$x^2 + \frac{b}{a}x = -\frac{c}{a}$$

Add $\left(\frac{b}{2a}\right)^2$ to both sides to complete the square:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Taking the square root and solving for x, we arrive at the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Part II: Pure Tufte Stream

Heading Huge (Tufte Measure)

Sub-Heading LARGE

Sub-Sub-Heading Large

Sub-Sub-Sub-Heading Large

In this section, text is strictly constrained to the optimal reading measure (31.2em $\approx$ 26pc), leaving the right margin open for annotations and side commentary.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Tufte Topic Topic environments adapt seamlessly to the narrower column width. The elastic label spacing ensures no overfull boxes occur.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

Tufte Sub-Topic Subtopic environments indent properly according to the font size and continue inline.

Morbi luctus, wisi viverra faucibus pretium, nibh est placerat odio, nec commodo wisi enim eget quam. Quisque libero justo, consectetuer a, feugiat vitae, porttitor eu, libero. Suspendisse sed mauris vitae elit sollicitudin malesuada. Maecenas ultricies eros sit amet ante. Ut venenatis velit. Maecenas sed mi eget dui varius euismod. Phasellus aliquet volutpat odio. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque sit amet pede ac sem eleifend consectetuer. Nullam elementum, urna vel imperdiet sodales, elit ipsum pharetra ligula, ac pretium ante justo a nulla. Curabitur tristique arcu eu metus. Vestibulum lectus. Proin mauris. Proin eu nunc eu urna hendrerit faucibus. Aliquam auctor, pede consequat laoreet varius, eros tellus scelerisque quam, pellentesque hendrerit ipsum dolor sed augue. Nulla nec lacus.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi.

Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.

Part III: Mid-Page Layout Dynamics

Seamless Stream Transitions

We begin this page in the **Tufte layout stream**. This paragraph is formatted to 65 characters per line, adhering to optimal typographic measure. Notice that the margin note to the right sits cleanly inside the canvas boundaries calculated by `geometry`'s `includemp` solver without overflowing the physical sheet.

This is a margin note typeset neatly within the reserved margin note area.

Narrow Analysis Here is an analytical point discussed in the narrow measure. As soon as a mathematical proof requires extra horizontal breathing room, we can open a scoped `layoutfullwidth` environment without issuing a page break.

Wide Proof Block (Full-Width Environment)

The layout has expanded across the gutter and margin area mid-page. Paragraphs now flow across the entire horizontal text area.

Theorem (Cauchy–Schwarz Inequality): For all vectors $\mathbf{u}, \mathbf{v}$ in an inner product space, the following inequality holds:

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \cdot \langle \mathbf{v}, \mathbf{v} \rangle$$

Proof: If $\mathbf{v} = \mathbf{0}$, the relation holds trivially as $0 \leq 0$. Assume $\mathbf{v} \neq \mathbf{0}$. For any scalar $\lambda \in \mathbb{R}$, the non-negativity of the inner product norm guarantees:

$$0 \leq \|\mathbf{u} - \lambda \mathbf{v}\|^2 = \langle \mathbf{u} - \lambda \mathbf{v},\ \mathbf{u} - \lambda \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle - 2\lambda \langle \mathbf{u}, \mathbf{v} \rangle + \lambda^2 \langle \mathbf{v}, \mathbf{v} \rangle$$

Since this quadratic polynomial in λ is always non-negative, its discriminant Δ must be non-positive:

$$\Delta = (-2\langle \mathbf{u}, \mathbf{v} \rangle)^2 - 4(\langle \mathbf{v}, \mathbf{v} \rangle)(\langle \mathbf{u}, \mathbf{u} \rangle) \leq 0 \implies 4|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq 4\langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle$$

Dividing by 4 completes the proof. ■

Return to Narrow Column

Notice what happened above: the proof expanded across the page, and this sentence **immediately snapped back** to the narrow Tufte measure on the **very same page**!

We are back in Tufte mode, so margin notes function normally again.

There was no page ejection, no vertical jump, and no layout corruption.

Scoped Safety The environment `layoutfullwidth` saved the previous layout stream and restored it upon exit. Furthermore, the `\marginpar` call inside the full-width block was suppressed, generating a clean warning in your compile log instead of printing over the edge of the paper.

Suspendisse vitae elit. Aliquam arcu neque, ornare in, ullamcorper quis, commodo eu, libero. Fusce sagittis erat at erat tristique mollis. Maecenas sapien libero, molestie et, lobortis in, sodales eget, dui. Morbi ultrices rutrum lorem. Nam elementum ullamcorper leo. Morbi dui. Aliquam sagittis. Nunc placerat. Pellentesque tristique sodales est. Maecenas imperdiet lacinia velit. Cras non urna. Morbi eros pede, suscipit ac, varius vel, egestas non, eros. Praesent malesuada, diam id pretium elementum, eros sem dictum tortor, vel consectetuer odio sem sed wisi.

Declarative Switch to Full-Width

You can also use declarative commands (`\setlayoutfullwidth` and `\setlayouttufte`) when a transition should persist indefinitely rather than inside a local environment block.

1. **Base Case ($n = 1$):** The left-hand side equals 1, and the right-hand side gives $\frac{1(1+1)}{2} = 1$. The base case holds.
2. **Inductive Hypothesis:** Assume the proposition holds for $k \geq 1$:

$$\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$$

3. **Inductive Step:** We examine the sum for $k + 1$:

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2}$$

   By induction, the equality holds for all natural numbers $n \in \mathbb{N}$. ■

Back to Tufte

This final paragraph is returned to the narrow reading measure by calling `\setlayouttufte`.

Final margin note verifying layout stability.

Sed feugiat. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Ut pellentesque augue sed urna. Vestibulum diam eros, fringilla et, consectetuer eu, nonummy id, sapien. Nullam at lectus. In sagittis ultrices mauris. Curabitur malesuada erat sit amet massa. Fusce blandit. Aliquam erat volutpat. Aliquam euismod. Aenean vel lectus. Nunc imperdiet justo nec dolor.